

Applied SVAR Identification

Day 2: Identification I — Zero Restrictions

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Course Roadmap

Day	Topic
1	VAR Foundations: Motivation, Estimation, Specification, Identification & Tools
2	Identification I: Short-Run Restrictions
3	Identification II: Long-run Restrictions
4	Identification III: Sign, Magnitude, and Narrative Restrictions
5	Identification IV: Instruments

Learning Objectives

By the end of today, you should be able to:

1. Explain the identification problem and the role of zero restrictions in solving it.
2. Explain when zero restrictions are appropriate and when they are not.
3. Explain how the Cholesky decomposition provides a solution via short-run zero restrictions.
4. Implement recursive (Cholesky) identification in MATLAB using the ACB Toolbox.
5. Apply short-run restrictions to partially replicate [Kilian \(2009\)](#) and [Wong \(2015\)](#).

Part 1

The Identification Problem

How can we recover the SVAR from the VAR?

VAR and SVAR

The **structural VAR** (SVAR) links the data to economically meaningful shocks:

$$B_0 y_t = b + \sum_{j=1}^p B_j y_{t-j} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, I_n)$$

The **reduced-form VAR** (VAR) we can estimate:

$$y_t = c + \sum_{j=1}^p A_j y_{t-j} + e_t, \quad e_t \sim \mathcal{N}(0, \Sigma)$$

where the connection is that VAR parameters are functions of the SVAR parameters

- ▶ Intercept: $c = B_0^{-1}b$
- ▶ VAR coefficient matrices: $A_j = B_0^{-1}B_j, j = 1, \dots, p$
- ▶ Error terms: $e_t = B_0^{-1}\varepsilon_t$
- ▶ Covariance matrix: $\Sigma = B_0^{-1}(B_0^{-1})'$

Identification Problem

Objective: Estimate the RFVAR and then recover the SVAR.

Identification Problem: The SVAR is 'underidentified' because:

1. Structural impact matrix, B_0^{-1} , has n^2 contemporaneous parameters.
2. Reduced form covariance matrix, Σ , has $\frac{n(n+1)}{2}$ unique parameters.

Recovering the SVAR therefore requires identifying assumptions.

Part 2

Short-Run Zero Restrictions

Restricting the contemporaneous impact of structural shocks

The Order Condition

The Order Condition

Since the impact matrix, B_0^{-1} , has n^2 unique elements, while the reduced form covariance matrix, Σ , has $\frac{n(n+1)}{2}$, unique parameters we can achieve exact identification by restricting $n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}$ elements of B_0^{-1} to be zero.

Comments:

- ▶ The order condition is met by restricting values to any constant, not just zero.
- ▶ The order condition is mathematical. It doesn't state which elements in the SVAR need to be restricted. That decision is up to us!

Short-Run Zero Restrictions

Short-Run Zero Restriction

A **short-run zero restriction** sets an element of B_0^{-1} to zero:

$$[B_0^{-1}]_{ij} = 0$$

so that structural shock j , ε_{jt} , has no contemporaneous effect on variable i , y_{it} .

Comments:

- ▶ The order condition states that we need to make $\frac{n(n-1)}{2}$ such restrictions.
- ▶ Given Sims' critique of DSEMs, such restriction should be motivated by economic theory.

Short-Run Recursive Identification

Sims (1980) proposed a **short-run recursive ordering** of the variables in y_t , which amounts to restricting the upper triangle of B_0^{-1} to zero:

$$B_0^{-1} = \begin{bmatrix} * & 0 & 0 \\ * & * & 0 \\ * & * & * \end{bmatrix}$$

Natural for three reasons:

- ▶ Order condition satisfied: There are $\frac{n(n-1)}{2}$ elements above the main diagonal
- ▶ Unique mapping: Obtained via the **Cholesky decomposition**
- ▶ Can (sometimes!) be motivated by economic theory — need to be careful here!

Cholesky Decomposition

Cholesky Decomposition

Any *positive definite symmetric matrix*, Σ , can be factorized as:

$$\Sigma = PP'$$

where P is a *lower triangular matrix* with positive elements on the main diagonal, called the (*lower*) *Cholesky factor* of Σ , and P' is its *conjugate transpose*.

Comments:

1. Useful because every covariance matrix is a positive definite symmetric matrix, implying that they all have a unique Cholesky factor.
2. Identification achieved by setting $B_0^{-1} = P$.
3. Named after French mathematician [André-Louis Cholesky](#).
4. In the ACB Toolbox, this is handled by a single call: `'VARopt.ident = 'short';`

Stock and Waston (2001): Recursive Ordering

Stock and Watson (2001) assume a recursive ordering:

$$y_t = (\pi_t, u_t, R_t)'$$

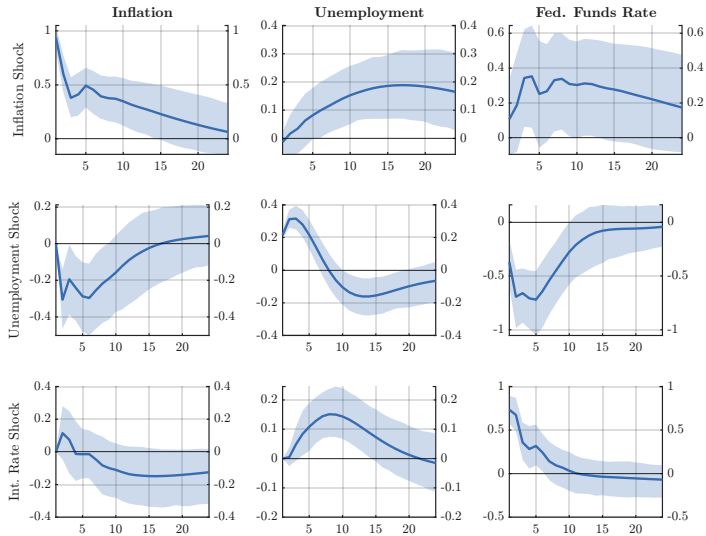
Economic interpretation of the restrictions:

1. **Inflation** (π_t) does not respond contemporaneously to unemployment or interest rate shocks. Prices are *sticky*: they are predetermined within a quarter.
2. **Unemployment** (u_t) does not respond contemporaneously to interest rate shocks. Labour market adjustment takes at least one quarter.
3. **Fed funds rate** (R_t) can respond contemporaneously to inflation and unemployment: the Fed reacts to current conditions within a quarter.

Policy Lens

The identification strategy embeds a *theory of monetary transmission*: policymakers react to changes in inflation and output quickly, however monetary policy affects inflation and unemployment with a lag.

SW (2001): IRFs with 68% CI over 24 quarters.



Ex: Interpret the results

Monetary policy in SVARs

- ▶ Standard practice in VAR analysis is to identify the monetary policy shock, $\varepsilon_{MP,t}$, from a monetary policy feedback rule, f .
- ▶ Consider the monetary policy reaction function

$$R_t = f(I_t) + \sigma_R \varepsilon_{R,t},$$

where

1. R_t is the monetary policy instrument (e.g. overnight bank rate)
 2. I_t is the set of available information at date t
 3. f is the monetary authorities feedback rule
 4. $\varepsilon_{R,t}$ is the monetary policy shock with standard deviation σ_R
- ▶ The reaction function contains variables that the central bank focus on, e.g. unemployment and inflation – sometimes called **systematic monetary policy**
 - ▶ The shock measures deviations from the reaction function – sometimes called **non-systematic monetary policy**

Common monetary policy reaction function

Stock and Watson's (2001) SVAR is based on a *Taylor rule* in which the behavior of the central bank is modeled as setting the interest rate based on present and past rates of inflation and unemployment:

$$R_t = R^* + \beta_1(\bar{\pi}_t - \pi^*) + \beta_2(\bar{u}_t - u^*) + \text{lagged}\{\pi, u, R\} + \varepsilon_{i,t},$$

where

1. R^* , π^* and u^* are target values of the real rate of interest, inflation and unemployment (end up in the constant)
2. $\bar{\pi}_t$ and $\bar{u}_t$ are the average values of inflation and unemployment rate over the past four quarters

The policy rule can be expanded by adding additional variables to the SVAR, changing the functional form, etc, (see e.g., [Sims and Zha \(2006\)](#) for markov-switching or [Primiceri \(2005\)](#) for time-varying parameters).

Ordering Variables: Economic Principles

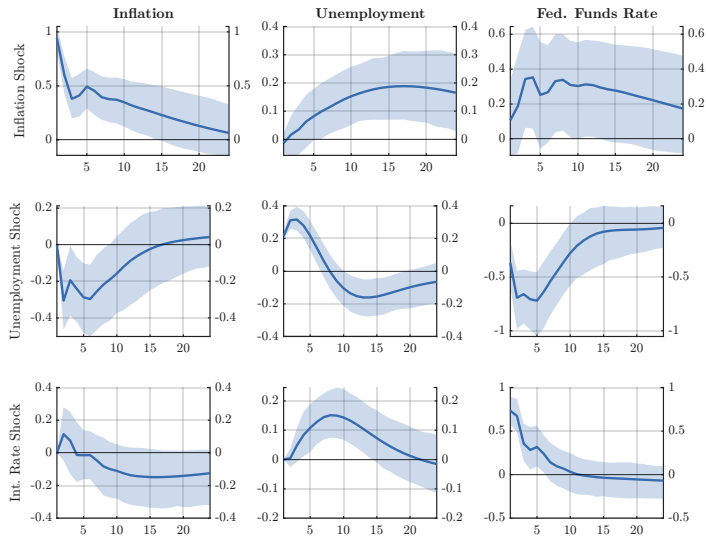
Christiano, Eichenbaum, and Evans (1998, proposition 4.1) show that the even though the recursiveness assumption is often insufficient to identify all the elements of the impact matrix, it can still reliably identify monetary policy shocks.

Building further, in Christiano, Eichenbaum, and Evans (2005, p.15) they suggest ordering VARs as $y_t = (y_{1t}, S_t, y_{2t})'$:

1. **Slow variables first:** Quantities, like output, employment, etc.
2. **Policy variable(s) in the middle:** policymakers observe and react to economic conditions within the period.
3. **Fast variables at the end:** Financial variables like exchange rates, equities, bonds, etc.

Ex: Re-estimate the Stock and Watson (2001) model with inflation and unemployment in the reverse ordering. Do the effects of monetary policy shocks change?

SW (2001) IRFs with $y_t = (u_t, \pi_t, R_t)'$



Ex: Do the effects of monetary policy shocks change?

Short-Run Restrictions: Pros and Cons

Pros

- ▶ Works for any VAR model
- ▶ Simple to implement and interpret
- ▶ Uniquely identified — no ambiguity
- ▶ Widely used → easy to benchmark
- ▶ Computationally fast

Cons

- ▶ Strong assumption
- ▶ Economic theory may not support full exogeneity → restrictions can be arbitrary
- ▶ Can be sensitive to the ordering

Key Idea

The key question: Is there a theoretically motivated recursive ordering? If yes, recursive identification is a natural starting point. If not, consider other approaches.

Part 2

Kilian (2009)

“Are the effects of oil market shocks on the price of oil all alike?”

Kilian (2009): Overview

Question: Are oil market demand or supply shocks more important drivers of the price of oil?

Motivation: The literature treated oil price shocks as a single phenomenon. But the cause of the shock matters:

- ▶ A supply disruption (e.g., OPEC cut) raises prices by restricting supply.
- ▶ A surge in global demand (e.g., 2003–2008) raises prices by increasing demand.
- ▶ Precautionary demand (fear of future scarcity) also raises prices.

Contribution: A 3-variable SVAR that *disentangles* three distinct structural shocks driving the real price of crude oil.

Identification: Short-run zero restrictions, with recursive ordering motivated by the structure of global oil markets.

Kilian (2009): Oil Market SVAR

- ▶ Oil Market SVAR: 1973:M1–2007:M12, $p = 24$ lags.
- ▶ Three variables in recursive order:
 1. q_t = Percentage change in global crude oil production
 2. y_t = Kilian index of real global economic activity
 3. p_t = Log of real price of oil
- ▶ Three structural shocks:
 1. Oil supply shock: unanticipated disruption to global oil production
 2. Aggregate demand shock: unexpected change in global real economic activity
 3. Oil-specific demand shock: Shifts in precautionary demand for oil

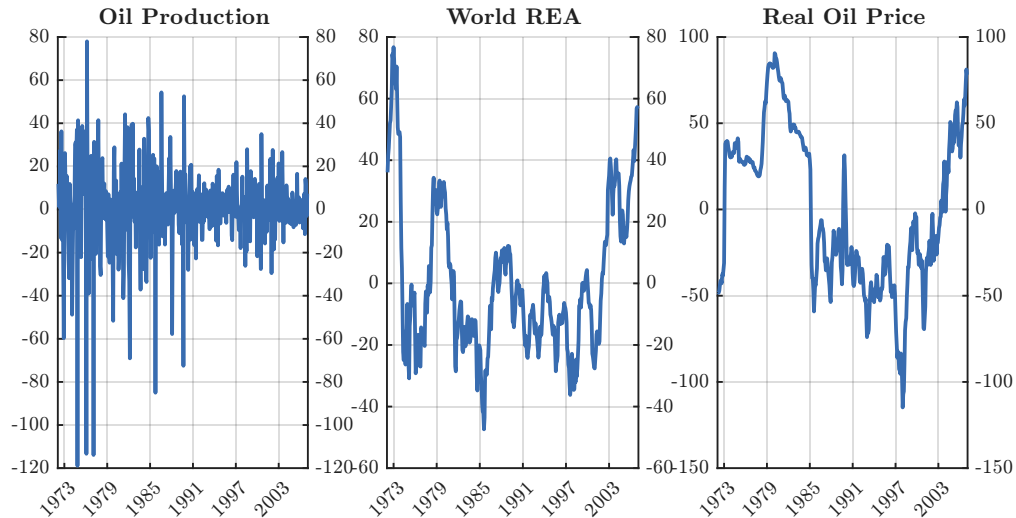
Kilian (2009): The Recursive Ordering

Economic justification for recursive ordering:

1. Oil supply curve is perfectly inelastic within the month $\rightarrow$ global oil production does not respond contemporaneously to other variables in the system $\rightarrow$
 $[B_0^{-1}]_{12} = [B_0^{-1}]_{13} = 0$
2. Global real economic activity does not respond to oil real oil price changes within the month $\rightarrow [B_0^{-1}]_{23} = 0$:

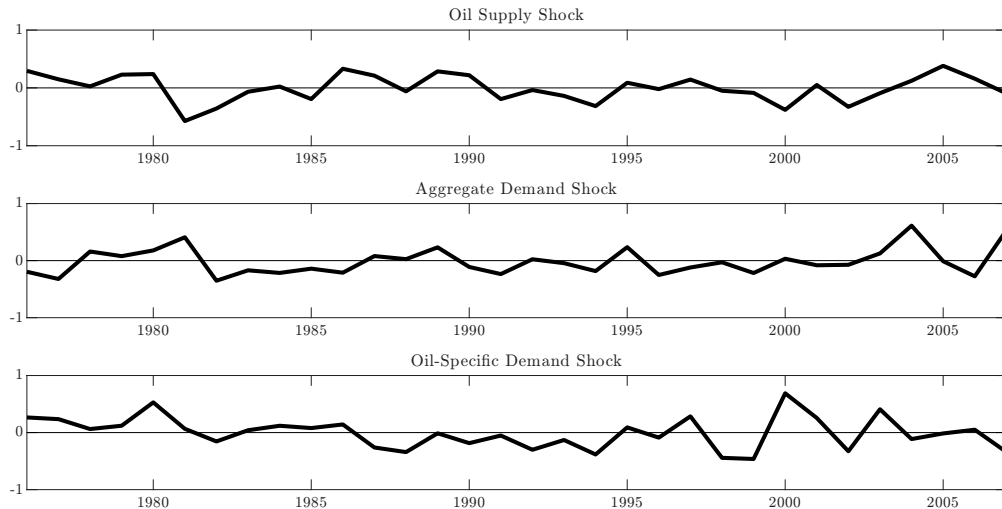
Note: These assumptions have since been refined by a huge literature on the use of SVARs for modeling oil markets.

Kilian (2009): Data, 1973–2007



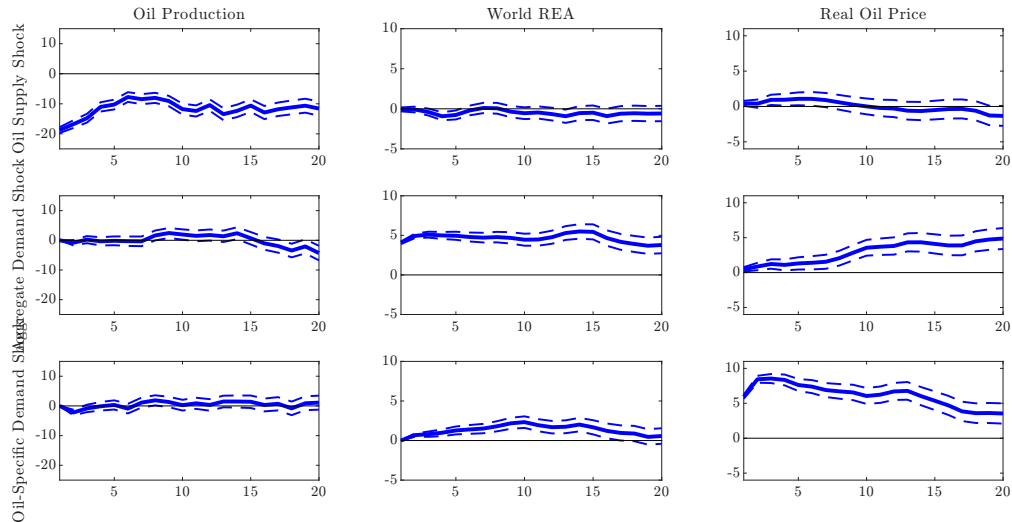
Ex: What key episodes in global oil markets are visible in the data?

Kilian (2009): Historical Structural Shocks, 1976–2007



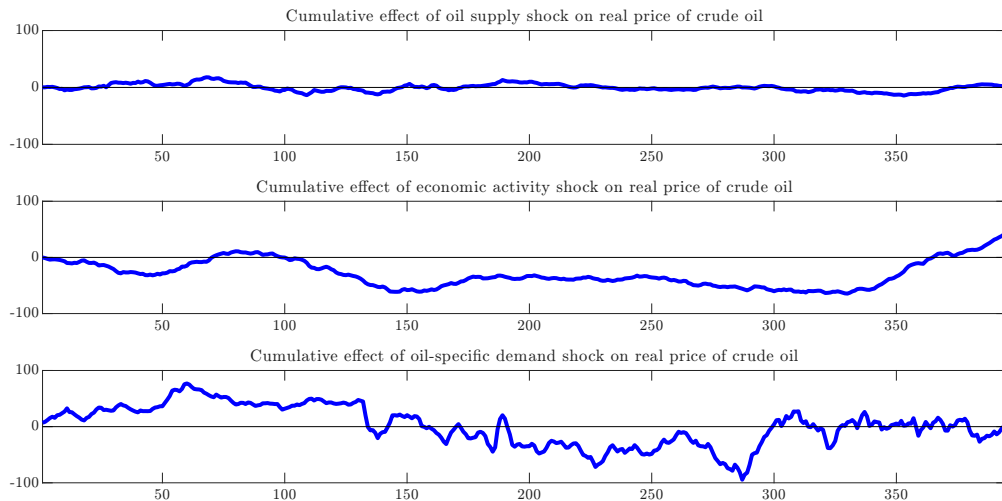
Shocks align with knowledge of oil markets: supply shock spikes in 1973–74 and 1990, and that his demand shock rises through the 2000s (seeds of narrative identification?)

Kilian (2009): Impulse Responses to Each Structural Shock



Ex: What are the dynamic effects of the shocks on the real price of oil?

Kilian (2009): Historical Decomposition of the Real Oil Price



Ex: What drove the 2003–2008 surge in oil prices?

Key Takeaways: Kilian (2009)

1. **Shocks differ:** Oil supply shocks, aggregate demand shocks, and oil-specific demand shocks have qualitatively and quantitatively different effects on real oil prices and the macroeconomy.
2. **Demand drove the 2003–2008 surge:** The historical decomposition shows that the 2000s oil price boom was predominantly driven by *global demand* — not supply disruptions.
3. **Supply shocks are short-lived:** Oil supply disruptions raise the real oil price only temporarily; their macroeconomic effects are modest and quickly reversed.
4. **Policy implication:** The appropriate policy response to an oil price shock depends on its source. A demand-driven surge calls for a different response than a supply disruption.

Part 3

Wong (2015)

“Do inflation expectations propagate the inflationary impact of oil price shocks?”

Wong (2015, JMCB): Overview

Question: Do oil price shocks that shift inflation expectations produce larger and more persistent effects on actual inflation?

Motivation: Standard VARs treat oil price pass-through to CPI as a direct mechanical effect. But *expectations* may amplify the transmission:

Oil price $\uparrow \rightarrow$ Inflation expectations $\uparrow \rightarrow$ CPI inflation $\uparrow$

If long-run inflation expectations are well-anchored, the second arrow is weak.

Contribution: Use Michigan Survey expectations to isolate the role of the expectations channel in transmitting oil price shocks to CPI inflation.

Identification: Short-run zero restrictions, with recursive ordering motivated by information difference between oil prices, expected inflation and actual inflation within the month.

Wong (2015): The VAR

- ▶ Two monthly VAR specifications (estimated separately), $p = 12$ lags:
 1. 1yr baseline: 1983:M1–2015:M3 (Michigan Survey 1yr-ahead expectations)
 2. 5-10yr baseline: 1990:M4–2015:M3 (Michigan Survey 5-10yr-ahead expectations)

- ▶ Three variables in recursive order:
 1. p_t = Log real price of oil
 2. π_t^e = Inflation expectations (1yr or 5–10yr ahead)
 3. π_t = CPI inflation (annualised month-on-month)

- ▶ Structural shock of interest:
 1. **Oil price shock**: an unanticipated increase in the real price of crude oil — does it shift inflation expectations and amplify pass-through to CPI?

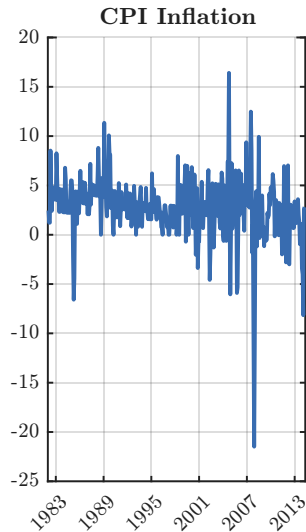
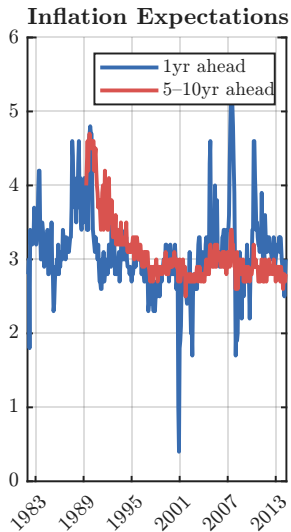
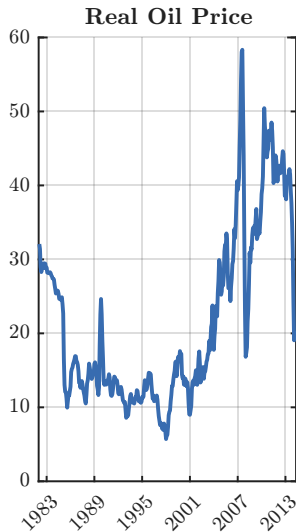
Wong (2015): The Recursive Ordering

Economic justification for recursive ordering (oil $\rightarrow$ expectations $\rightarrow$ CPI):

1. **Real oil price** does not respond contemporaneously to US inflation expectations or CPI $\rightarrow$ the real price of oil is determined on global commodity markets, not by US domestic conditions within a month $\rightarrow [B_0^{-1}]_{12} = [B_0^{-1}]_{13} = 0$
2. **Inflation expectations** respond to oil prices but are predetermined relative to realised CPI within the same month $\rightarrow$ survey respondents form expectations before the monthly CPI release $\rightarrow [B_0^{-1}]_{23} = 0$

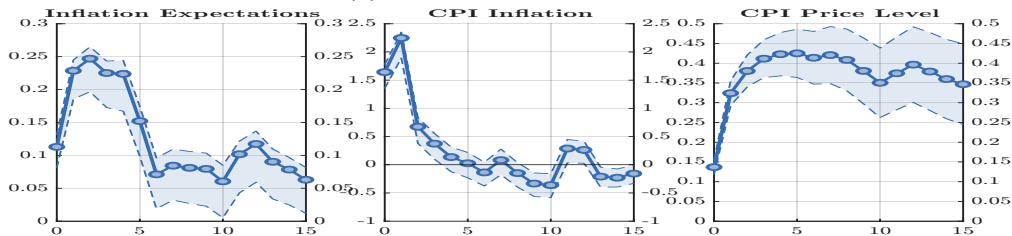
Note: This ordering parallels Kilian (2009) — the upstream variable (oil price) does not respond contemporaneously to downstream outcomes (expectations and inflation).

Wong (2015): Data, 1983–2015

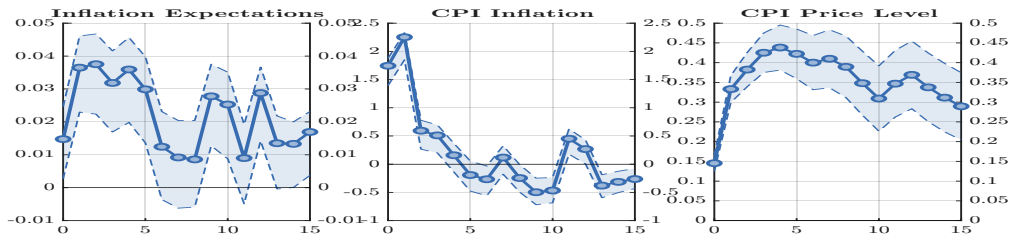


Wong (2015): Impulse Responses to a 10% Oil Price Shock

(a) 1yr ahead, 1983:1–2015:3

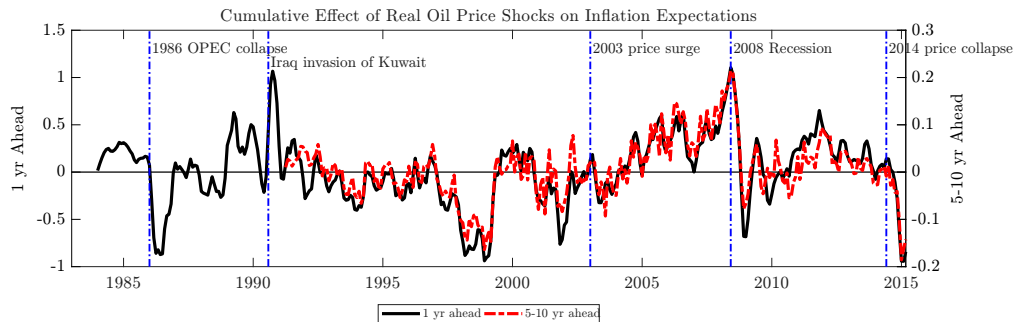


(b) 5–10yr ahead, 1990:4–2015:3



Ex: Interpret the results.

Wong (2015): Oil Price Contribution to Inflation Expectations



Ex: When do oil shocks most strongly shift expectations — 1yr or 5-10yr ahead?

Wong (2015): Counterfactual Analysis

Question: Does the sensitivity of inflation expectations to oil shocks translate into higher actual CPI inflation? i.e., are expectations a meaningful *transmission channel*?

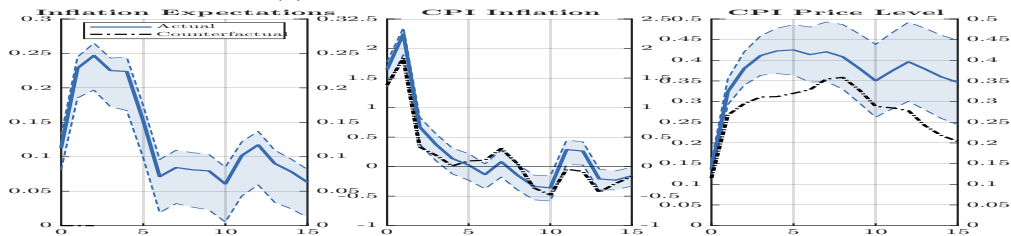
Approach: Following Sims and Zha (2006), construct a counterfactual IRF that *shuts down* the expectations channel:

1. At each horizon h , solve for the sequence of structural expectations shocks that zeros out the expectations response to the oil price shock.
2. Accumulate these shocks to obtain counterfactual paths for all variables.

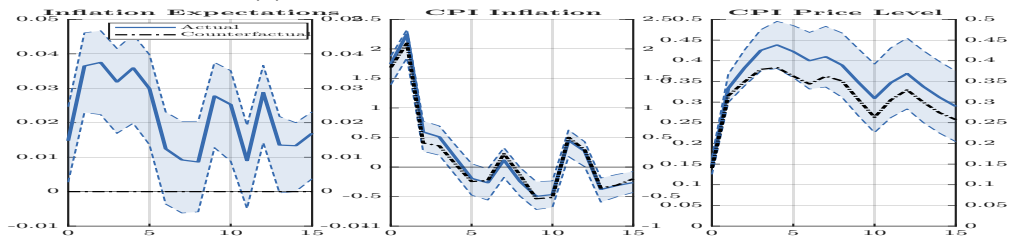
Interpretation: If the counterfactual CPI response (no expectations channel) matches the actual response, expectations add nothing to the transmission of oil shocks to inflation.

Wong (2015): Counterfactual IRFs

(a) 1yr ahead expectations, 1983:1–2015:3

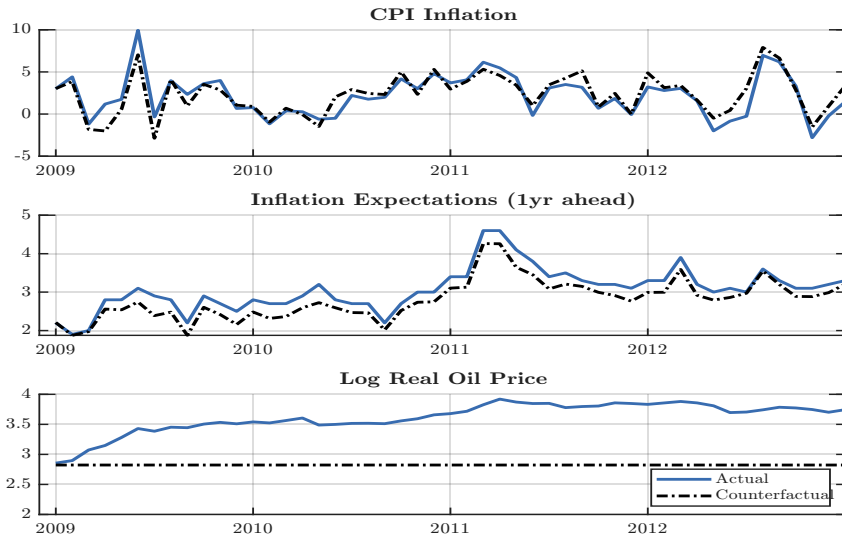


(b) 5–10yr ahead expectations, 1990:4–2015:3



Ex: Does shutting the expectations channel alter the inflation or price-level response?

Wong (2015): Counterfactual Great Recession Exercise



Ex: What does the counterfactual imply about the role of oil prices in the “missing deflation” of the Great Recession?

Key Takeaways: Wong (2015)

1. **Expectations are a channel:** U.S. inflation expectations are sensitive to real oil price shocks and may act as triggers for second-round inflation if inflation expectations materialize into higher actual inflation.
2. **Expectations move with oil prices:** Historical decomposition shows movements in the real oil price drive inflation expectations.
3. **Counterfactual results suggest they don't matter much:** movements in inflation expectations in response to real oil price shocks may have become irrelevant in understanding inflation dynamics at some point after the 1990s, and were not an explanation for the 'missing deflation' of the Great Recession.

Part 4

Aastveit, Bjornland and Cross (2023)

“Do inflation expectations and the associated pass-through of oil price shocks depend on demand and supply conditions underlying the global oil market?”

Aastveit, Bjornland and Cross (2023, ReStat): Overview

Question: Do oil market demand and supply shocks impact inflation expectations differently? If so, can these different impacts produce larger and more persistent effects on actual inflation?

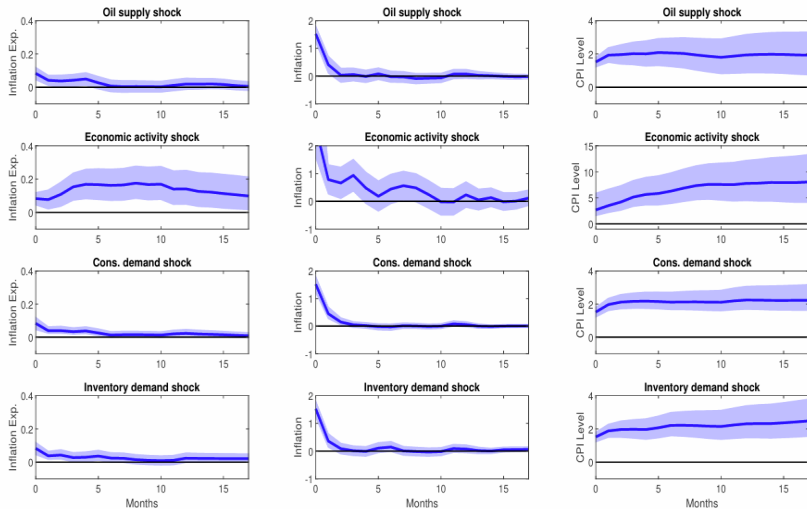
Motivation: Revisit Wong (2015) using insights from Kilian (2009) that not all oil price shocks are alike. Expectations may amplify the transmission:

Oil market demand/supply shock $\rightarrow$ Oil price $\uparrow \rightarrow$ Inflation expectations $\uparrow \rightarrow$ Inflation $\uparrow$

Contribution: Show that they do! Inflation expectations propagate the pass-through the price of oil to inflation when oil price responses are persistent (e.g., aggregate demand shock) as opposed to short-lived (e.g., oil supply shock). Also helps explain the missing deflation puzzle of the Great Recession.

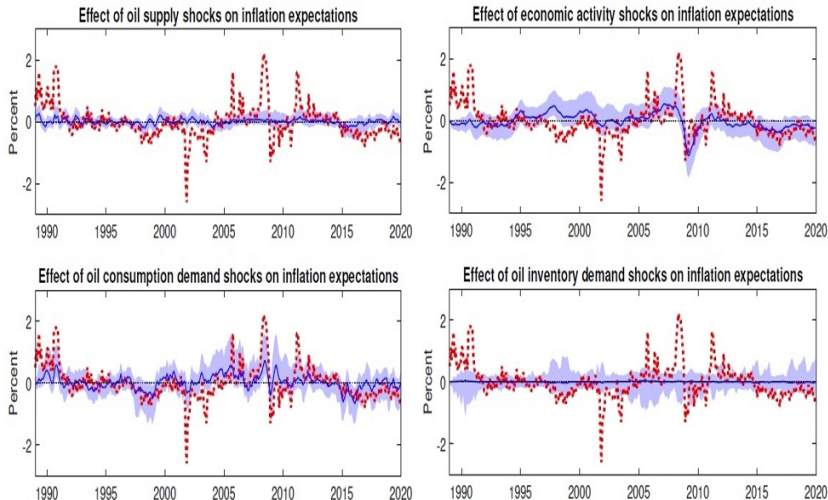
Identification: Zero, sign, and magnitude restrictions (Day 4).

ABC (2023): IRFs



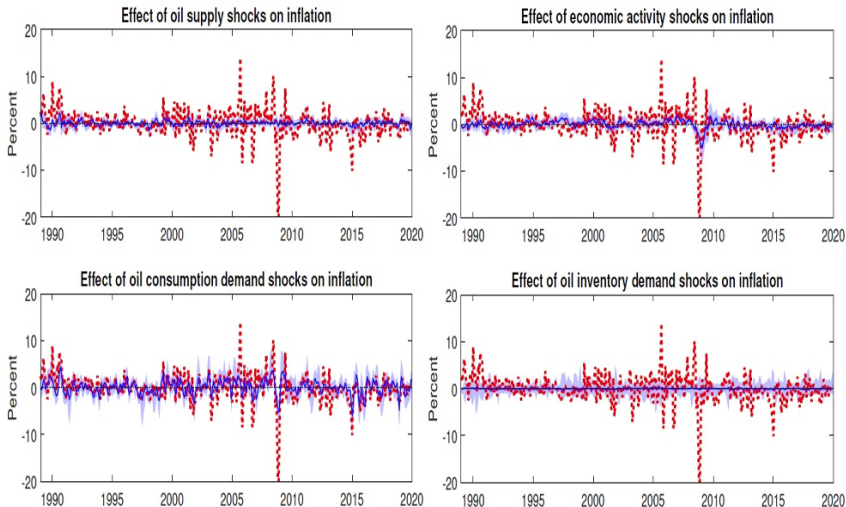
Ex: Interpret the results.

ABC (2023): HD Inflation Expectations



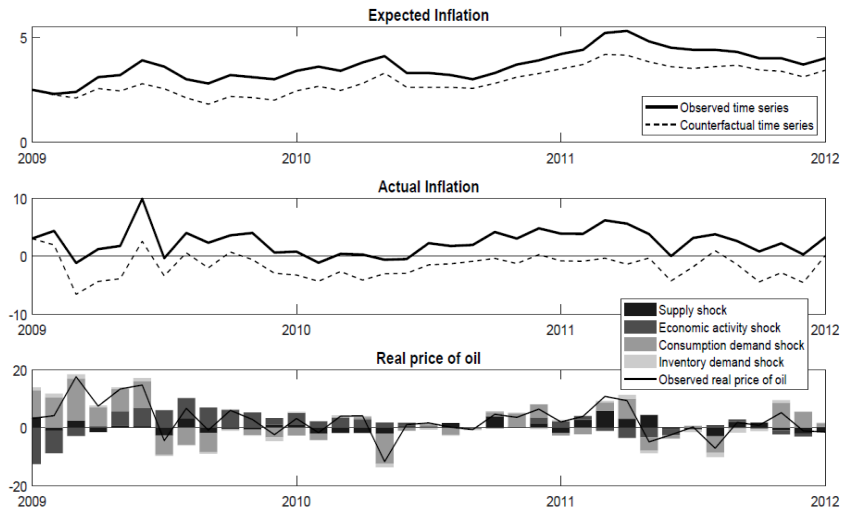
Ex: Interpret the results.

ABC (2023): HD Inflation Expectations



Ex: Interpret the results.

ABC (2023): Counterfactual



Ex: Interpret the results.

Main Takeaways

1. **Order Condition:** For the SVAR to be exactly identified we need $\frac{n(n-1)}{2}$ restrictions to be made.
2. **Recursive Identification:** Restrict the values of B_0^{-1} above the main diagonal to be zero $\rightarrow$ exactly $\frac{n(n-1)}{2}$ restrictions! But need to justify it using theory.
3. **Cholesky Decomposition:** A way to decompose the covariance matrix into a lower triangular matrix that we can use to identify the impact matrix.
4. **Partial Identification:** Even if recursive identification can't reliably fully identify each of the shocks, it may partially identify a specific shock of interest, e.g., monetary policy shock.
5. **Counterfactual Analysis:** A scenario where we investigate what would have happened if the world was different, e.g., inflation expectations did not respond to oil prices.

Preview of Day 3

Long-run (zero) restrictions — which restricts the *long-run* effect of a shock to zero rather than its *contemporaneous* effect, with applications to

1. Output and unemployment (Blanchard and Quah, 1989)
2. Monetary policy and exchange rates (Bjørnland, 2009).

Thank you!

